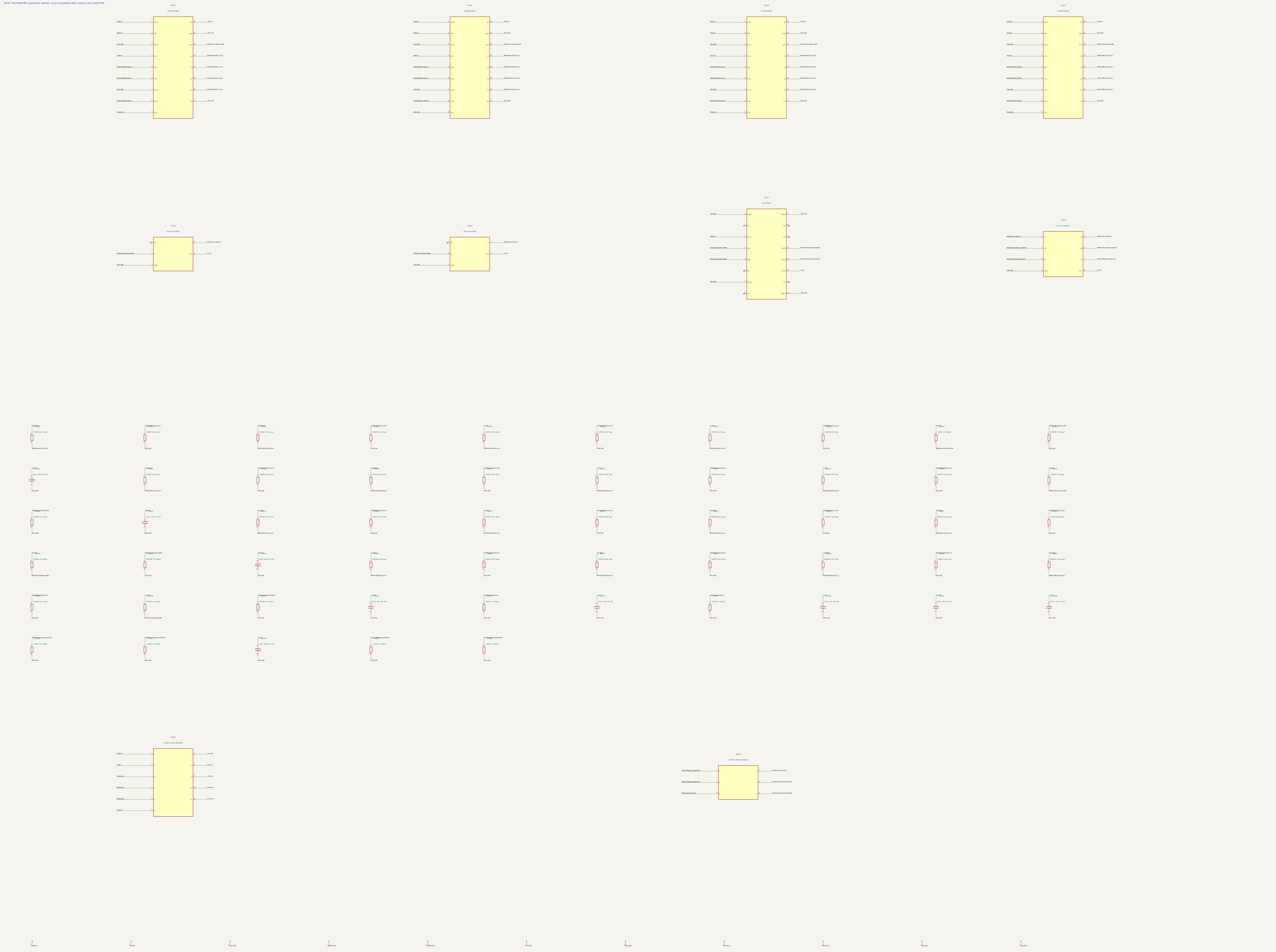
MOTOR POWER HEALTH / ONE MOTHERBOARD / SOURCE CANDIDATE r0.1

Four actual voltage—window monitors; FIELD/LOGIC remain isolated; no supply or limiter created here.

Health = receiving 5V and 3.3V valid AND isolated AUX5/AUX24 valid; no FPGA/software bypass.

LTC fault40us bound requires10% overdrive. ISO power1.7-2.25V undetermined. Physical brownout qualification remains open.

AUX24 inherited18-30V requirement retained; current guaranteed static arming screen only19-29V.


U9001
LTC2962HUD#PBF
CORE_5V 1 RDIS
CORE_5V 2 MR
LOGIC_GND 3 DVCC
CORE_5V 4 VCC
MOTOR_PWR_REF_LOGIC_A 5 REF
MOTOR_PWR_REF_LOGIC_A 6 PG1
LOGIC_GND 7 PG2
MOTOR_PWR_REF_LOGIC_A 8 PG3
LOGIC_GND 9 PG4
RT 10 CORE_5V
GND 11 LOGIC_GND
RST 12 MOTOR_LOGIC_HEALTH_A_RAW
V4 13 MOTOR_PWR_LOGIC_A_1_OV
V3 14 MOTOR_PWR_LOGIC_A_1_UV
V2 15 MOTOR_PWR_LOGIC_A_0_OV
V1 16 MOTOR_PWR_LOGIC_A_0_UV
EP 17 LOGIC_GND
U9005
SN74LVC1G17DBVR
1 NC
MOTOR_LOGIC_HEALTH_A_RAW 2 A
LOGIC_GND 3 GND
Y 4 MOTOR_LOGIC_HEALTH_A
VCC 5 IO_3V3
U9007
ISO7720FDWR
FIELD_GND 1 GND1
2 NC
FIELD_5V 3 VCC1
MOTOR_FIELD_HEALTH_A_RAW 4 INA
MOTOR_FIELD_HEALTH_B_RAW 5 INB
6 NC
FIELD_GND 7 GND1
8 NC
GND2 9 LOGIC_GND
NC 10
NC 11
OUTB 12 MOTOR_FIELD_HEALTH_B_ISOLATED
OUTA 13 MOTOR_FIELD_HEALTH_A_ISOLATED
VCC2 14 IO_3V3
NC 15
GND2 16 LOGIC_GND
U9008
SN74LVC2G08DCUR
MOTOR_LOGIC_HEALTH_A 1 1A
MOTOR_FIELD_HEALTH_A_ISOLATED 2 1B
MOTOR_POWER_OK_B_QUALIFIED 3 2Y
LOGIC_GND 4 GND
2A 5 MOTOR_LOGIC_HEALTH_B
2B 6 MOTOR_FIELD_HEALTH_B_ISOLATED
1Y 7 MOTOR_POWER_OK_A_QUALIFIED
VCC 8 IO_3V3
B9001
INTERNAL REVIEW BOUNDARY
CORE_5V 1
IO_3V3 2
LOGIC_GND 3
MOTOR_5V_A 4
MOTOR_5V_B 5
FIELD_5V 6
7 FIELD_GND
8 AUX_5V_A
9 AUX_5V_B
10 AUX_24V_A
11 AUX_24V_B
B9002
INTERNAL REVIEW BOUNDARY
MOTOR_POWER_OK_A_QUALIFIED 1
MOTOR_POWER_OK_B_QUALIFIED 2
MOTOR_LOGIC_HEALTH_A 3
4 MOTOR_LOGIC_HEALTH_B
5 MOTOR_FIELD_HEALTH_A_ISOLATED
6 MOTOR_FIELD_HEALTH_B_ISOLATED

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File: motherboard_motor_power.kicad_sch

Title: Motor receiving and field power health

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